Project Proposal

Customer Intelligence & Retention Platform (CIRP)

Project: Customer Intelligence & Retention Platform (CIRP)

Objective:

- Predict which customers will stop buying in the next 90 days, and explain why.
- Group customers into behavioural segments with a name and a recommended action.
- Show both results in one Streamlit app that ranks customers by priority and expected revenue at risk.

Description:

CIRP is a churn prediction and customer segmentation platform for a retail business. It uses the public Online Retail II dataset: 5 quarterly snapshots, 5,253 customers and 22 behavioural features (recency, frequency, monetary value, returns, trends). A customer is churned when they place no order in the 90 days after the snapshot. Data is split by time into train / validation / test to avoid leakage. A Random Forest predicts churn probability, K-Means groups customers into 6 segments, and both outputs are joined in a Customer 360 table with a priority score (0.6 x risk + 0.4 x value) and expected revenue at risk.

Scope: data preparation and EDA, churn model, segmentation model, explanations, Streamlit app (local) and documentation.

Out of scope for now: cloud deployment, real-time data, CRM integration and sending campaigns.

Tools & Design:

- Data layer: raw transactions -> cleaning -> customer features -> time-based split -> RobustScaler + feature selection -> data package.
- Model layer: Random Forest churn model + K-Means segmentation, with evaluation reports and saved joblib models.
- App layer: Streamlit app loads the saved models and shows churn risk, top factors (feature importance), segment and recommended action.
- Dashboard contract: the app receives a fixed 11-feature preset; cleaning and feature engineering stay in the Python backend.

Technologies to Use:

- Python, pandas, NumPy and Jupyter Notebook.
- scikit-learn 1.9: Logistic Regression, Decision Tree, Random Forest, K-Means, DBSCAN, RobustScaler.
- joblib for models; matplotlib and seaborn for charts and evaluation reports.
- Streamlit for the app; Git and GitHub for version control (one branch per member, pull requests).

Team Members & Responsibilities:

1- **Mohamed Atef Mohamed Abdo** - Data source & preprocessing (Project Leader) - data source, EDA, data dictionary, documentation and delivery.

2- **Mohamed Ahmed Ebrahim Mohamed** - Data source & preprocessing - cleaning, feature engineering, scaling and the saved preprocessor.

3- **Omar Diaa El-Sayed** - Segmentation - K-Means vs DBSCAN, number of segments, profiles and business names.

4- **Sohaib Mohamed Mohamed Mustafa** - Prediction - churn classifiers, tuning, threshold, explanations and evaluation.

5- **Mohamed Hamada Omar** - UI/UX - app pages, charts and the user journey in Streamlit.

6- **Hussein El-Desouky Ibrahim Othman** - Integration - load models, batch scoring, input validation and end-to-end testing.

Week 1 (2 - 8 Oct 2026): System analysis & design

- Mohamed Atef - system analysis & design document.
- Mohamed Hamada - wireframes and user journey.

- Hussein - model-app interface and Streamlit skeleton with a dummy model.
- Sohaib and Omar - experiment plan, algorithms and segmentation features.

Deliverables:

- System analysis & design document (8 Oct).
- Wireframes and interface contract.
- Running Streamlit skeleton.

Week 2 (9 - 15 Oct 2026): Baselines and app skeleton

- Sohaib - baseline churn models (Logistic Regression, Decision Tree, Random Forest) with Recall, F1 and PR-AUC.
- Omar - first K-Means run and K selection (elbow, silhouette).
- Mohamed Hamada and Hussein - multi-page Streamlit skeleton.
- Mohamed Ahmed - sample customers for the demo and the tests.

Deliverables:

- Baseline results table.
- K selection plots.
- Working multi-page skeleton and sample file.

Week 3 (16 - 22 Oct 2026): Tuning and app features

- Sohaib - tune the churn model, choose the threshold on validation and add feature-importance explanations.
- Omar - K-Means vs DBSCAN, 6 segments with names, descriptions and actions.
- Mohamed Hamada and Hussein - result views, segment page and model integration.
- Mohamed Ahmed - check that the 11-feature preset matches training.

Deliverables:

- Tuned model with evaluation report.
- 5 segment profiles.
- App pages connected to the saved models.

Week 4 (23 - 29 Oct 2026): Final models and delivery

- Sohaib and Omar - final training on full_train, one test evaluation and saved models (26 Oct).
- Hussein - batch scoring, input validation and end-to-end tests (27 Oct).
- Mohamed Hamada - UI polish, priority list and CSV export.
- Mohamed Atef - final report, README, presentation and GitHub upload (29 Oct).

Deliverables:

- Final models and preprocessor (26 Oct).
- Integrated app (28 Oct).
- Test report, final report and presentation (29 Oct).